Nalan Liyanage

Data Scientist

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Profile Summary

Computer Science graduate from the University of Colombo School of Computing with hands-on experience in machine learning, time-series forecasting, Generative AI, and production-oriented data systems. Built end-to-end ML pipelines, predictive and forecasting models, and LLM/RAG and agentic AI applications using Python, SQL, TensorFlow, Scikit-learn, LightGBM, and LangChain/LangGraph, with backend delivery in Flask, Express.js, and REST APIs. Strong in data preprocessing, feature engineering, model evaluation, and explainability (SHAP). Familiar with Azure Data Factory and Azure Databricks, and currently working with observability tooling to monitor AI systems. Eager to turn data into actionable business insights and deploy reliable, scalable AI solutions.

Education

University of Colombo School of Computing (UCSC) *Graduated: 2026*
BSc. in Computer Science
GPA: 3.5 / 4.0

Kingswood College, Kandy *Completed: 2021*
G.C.E. Advanced Level — Physical Science Stream
Results: Physics — A, Chemistry — A, Combined Mathematics — B

Experience

AI/ML Intern — Acentura *Aug 2026 – Present*

- Building an AI agent as a research project to automate a materials selection process in a manufacturing workflow, using LangGraph and PostgreSQL.
- Designing and implementing a Specification-Driven Development framework for AI agents, with governance and monitoring built on OpenTelemetry, Prometheus, Loki, and Grafana.
- *Tech stack:* Python, LangGraph, PostgreSQL, OpenTelemetry, Prometheus, Loki, Grafana.

Software Engineering Intern — Sezenta *2025*

- Completed a 6-month internship contributing to full-stack development across frontend and backend systems, working in an agile team using Git/GitHub for version control and collaboration.
- Built AI agent tools using LangChain, enabling AI agents to call backend services and perform tasks.
- Developed frontend features with React and Next.js and backend APIs with NestJS and TypeScript.
- *Tech stack:* TypeScript, React, Next.js, NestJS, LangChain.

Projects

1. **BloxLogicAI: AI & Blockchain-Enabled Supply Chain Forecasting System** — GitHub *2026*
   - Designed and built an end-to-end AI system for Sri Lanka's tea export industry, forecasting monthly market demand and recommending optimal export quantities using Facebook Prophet, achieving 8.66% MAPE (RMSE: 1865.09) on a 12-month held-out test set trained on 14+ years of historical export data.
   - Implemented anomaly detection with Scikit-learn's Isolation Forest to flag supply chain risks (fuel price spikes, severe export drops, crude oil shocks) with severity-based classification and action recommendations.
   - Engineered a custom Python blockchain simulation (SHA-256 hash-chaining) for tamper-proof traceability of tea batches across 5 supply chain stages, and built interactive User/Admin portals using Streamlit.
   - *Tech stack:* Python, Streamlit, Pandas, Facebook Prophet, Scikit-learn, SHA-256/Hashlib.

2. **Loan Default Risk & Approval Prediction System** — GitHub *2025*
   - Developed two ML pipelines: (1) LightGBM classification model to predict whether a loan will be approved or not, and (2) LightGBM regression model to generate a continuous loan risk score.
   - Performed preprocessing, feature engineering, handling missing values, and metric-driven evaluation.
   - Applied SHAP for model explainability and feature contribution analysis.
   - Built a Flask-based web interface for interactive loan approval and risk score predictions.
   - *Tech stack:* Python, LightGBM, SHAP, scikit-learn, Flask.

3. **TourMateAI: AI-Powered Travel Companion for Sri Lanka Tourism** — GitHub *2026*
   - Built a full-stack AI travel platform providing context-aware recommendations based on user preferences, location, and live weather, using a content-based recommendation engine (cosine similarity) in scikit-learn.
   - Developed a RAG chatbot with Google Gemini, ChromaDB, and sentence-transformers, and a multi-agent itinerary planner using LangGraph and LangChain to generate optimized travel schedules.
   - Trained a custom CNN for Sri Lankan landmark image recognition, and added multi-language support.
   - Designed a three-tier architecture with a Flask modular backend, MySQL, Firebase authentication, and a React frontend.
   - *Tech stack:* React, Python, Flask, TensorFlow/Keras, Scikit-learn, LangGraph, LangChain, Gemini, ChromaDB, MySQL, Firebase, Google Maps & OpenWeather APIs.

4. **Medical Chatbot with Modular RAG Pipeline** — GitHub *2025*
   - Developed a complete RAG-based medical Q&A system using LangChain and Hugging Face.
   - Implemented data ingestion, text splitting, embedding generation, and vector indexing using Pinecone.
   - Built backend using Flask and Docker for scalable deployment.
   - *Tech stack:* Python, LangChain, Hugging Face, Pinecone, Flask, Docker.

5. **QuickAI: AI Content Generation SaaS Platform** — GitHub *2025*
   - AI SaaS platform where users generate text/image content from prompts, manage creations, and publish to a public feed with a like system.
   - Built an Express.js backend proxying OpenAI, Gemini, and Clipdrop APIs; integrated Clerk for authentication, Cloudinary for image storage, and a Neon PostgreSQL schema for the feed.
   - *Tech stack:* React, Tailwind CSS, Node.js, Express.js, Neon PostgreSQL, Clerk, Cloudinary, Gemini, Clipdrop.

Certifications

Mastering Data Science, ML, Deep Learning NLP *Ongoing (99 hrs)*
Full-Stack Web Development Bootcamp — Udemy *2025*

Technical Skills

Programming Languages: Python, SQL, JavaScript, TypeScript, R
Machine Learning & Data Science: scikit-learn, LightGBM, TensorFlow/Keras, Prophet, SHAP, pandas, NumPy, Matplotlib, Seaborn
Data Engineering Tools: ETL pipelines, Azure Data Factory, Azure Databricks
Databases: PostgreSQL, MySQL, Azure SQL, MongoDB, Vector Databases (Pinecone, Chroma)
Cloud & Development Tools: Git, GitHub, Docker, Azure, AWS (basic), Jupyter Notebook
Generative & Agentic AI: prompt engineering, LangChain, LangGraph, LangSmith, MCP, N8N
Monitoring & Observability: OpenTelemetry, Prometheus, Loki, Grafana

Soft Skills

Analytical Thinking — Problem Solving — Clear Communication — Adaptability — Cross-Functional Collaboration